

Executive Summary

Technology Partnership Opportunity

Urban Rig Europe invites institutional investors to co-develop the first European deployment of Urban Rig — a patented, continuous superheated-steam pyrolysis system that converts mixed, unsorted waste into four marketable oil fractions, biochar, and recoverable metals, with zero dioxin emissions. Urban Rig Europe is the regional arm for sales and programme management, with OneWorld Corporation (Osaka) as the R&D headquarters. The investor partners directly with the operational team that owns and manufactures the technology.

The partnership model is open by design. Site, feedstock strategy, scale, operating team, and capital structure are the investor's design space. We bring the technology, the IP, and a configurable financial model. The one non-negotiable: Urban Rig remains the technology partner and equipment supplier.

Proven Technology

Urban Rig operates continuously at 600°C in an oxygen-free environment, cracking organic polymers into hydrocarbons. Tested at industrial scale by Asada Shokai (Noda, Japan, 2022): oil yields of 70–82% from pure plastics across nine resin types, emissions 5–50x below French standards (KANISO Technos verified), oil quality certified by NKKK at 43.27 MJ/kg. Processes PVC and all halogenated plastics. The V2 model uses purchased heating oil for operational reliability. All char and off-gas by-products are sold commercially or monitored for process understanding. Modular: ships in 7 truck sections. A URC-2000 (200 T/day nominal capacity, ~140 T/day actual weight for MSW) runs with ~15 staff.

Illustrative Economics (URC-2000, Mixed MSW, France Base Case)

Machine nominal capacity: 200 T/day volumetric (140 T/day actual weight at 0.7 T/m³ MSW bulk density). Annual throughput: 46,200 T/year (330 operating days).

CAPEX: €63.8M.

Annual Revenue: €16.1M (gate fees 33% + oil 39% + char/biochar 9% + metals 4% + carbon credits 15%).

EBITDA: €8.9M (55% margin). Purchased heating oil ensures operational reliability. All char/biochar sold as commercial product.

Payback: ~7 years (simple); equity payback ~12 years.

These figures represent one output of a configurable model.

Inputs — feedstock composition, tipping fees, oil price (€480–895/T across three scenarios), geography, and plant scale — are fully adjustable. An interactive online simulator with AI-augmented regional scenario generation is in development.

Why Now

EU Circular Economy mandates 65% recycling and max 10% landfilling by 2035 — targets no member state currently meets. France incinerates 38% and landfills 18% of MSW. ICIS plastic pyrolysis oil reached €670/T in 2024–25 with growing ISCC-certified premiums. Urban Rig is not a concept: it is manufactured, tested, verified, and ready for deployment.

Engagement Process

Discovery (2-4 weeks) → Joint scenario development (4-8 weeks) → Independent technical validation, commissioned jointly (2-4 months) → Commercial structuring and financial close → Equipment manufacture and commissioning (12-18 months to operations).

The Problem

The global waste crisis is accelerating. The World Bank estimates 2.01 billion tonnes of MSW were generated worldwide in 2016, projected to reach 3.4 billion tonnes by 2050. In Europe, Eurostat reports 225 million tonnes in 2021 (513 kg per capita), with only 49.6% recycled or composted, 27.5% incinerated, and 22.9% still landfilled.

Source: World Bank “What a Waste 2.0” (2018); Eurostat Municipal Waste Statistics (2023).

The French Context

France generates 38.5 million tonnes of MSW annually (580 kg per capita, above the EU-27 average of 505 kg). Treatment relies heavily on incineration (38.1%) and landfilling (18.1%), with recycling at only 26.9%.

Source: ADEME 2022; Ministry of Ecological Transition (Ministry of Ecological Transition).

The Île-de-France region alone produces 6.2 million tonnes per year (17,000 tonnes/day), with 37% going to incineration and 18% to landfill. The EU Circular Economy Package mandates 65% recycling by 2035 and a maximum of 10% landfilling — targets France is far from meeting.

Source: ORDIF (Île-de-France Regional Waste Observatory) regional statistics; EU Directive 2018/851.

The Limits of Current Solutions

Conventional incineration generates dioxins, CO₂, and various atmospheric pollutants. It produces toxic bottom ash (machefer - boiler slag) and fly ash that require controlled landfilling, destroying material value in the process. Material recovery from waste-to-energy incineration is only 5–10%.

Mechanical recycling is limited: 76% of plastics in French MSW are either not recycled or have very low recyclability. PP has a 90% non-recycling rate, PS 95%, and composite plastics 100%. Even PET, the most recycled plastic, sees 34% go to incineration or landfill.

Source: Annex A — Current Status Analysis.

Safe incineration plants are prohibitively expensive and large-scale. Current processing relies on tipping fees alone with no resource recovery revenue, making waste management a pure cost centre.

The Solution: Urban Rig Technology

Urban Rig is a patented superheated-steam pyrolysis and carbonization system that transforms the waste management paradigm from cost centre to profit centre. Operating in a completely anoxic (oxygen-free) reactor environment, it converts mixed, unsorted waste directly into four marketable oil fractions, industrial-grade charcoal (carbide), and recoverable metals and glass.

Source: OneWorld Corporation technical documentation; Urban Rig product brochures.

Core Process Advantages

No Sorting Required — Direct Batch Input

Urban Rig does not require waste to be sorted, cleaned, or pre-treated before processing. Mixed waste — including mud, non-resin materials, metals, pesticides, and food residue — is loaded directly into transport buckets. The only preparation is one day of storage to reduce surface moisture. Superheated ultra-fine steam vaporizes and carbonizes all organic content while metals are extracted in a non-oxidized state, preserving their purity.

Source: OneWorld Corporation, Urban Rig Recycling Solutions ENG, p.2; Urban Rig ENG, p.2.

Handles Any Waste Condition

The system processes muddy, wet, and contaminated feedstock directly. This includes general household waste, food residue with high moisture content, sludge, soil, and agricultural waste. The URC-2000 is specifically engineered for carbonizing waste with high moisture content, a challenge that conventional pyrolysis systems cannot handle.

Source: UR2000-JP technical specification, p.2; Urban Rig ENG, p.4.

Mixed PVC and All Halogenated Plastics

Urban Rig processes multicomponent materials including crimped film, PVC, vinyl, and mixed resins, extracting high-quality oxygen-free recycled oil even from PVC waste. The catalytic layer within the reactor adsorbs chlorine, while steam neutralises hydrochloric acid that would otherwise corrode equipment. Latest testing demonstrates that this capability extends to all halogenated compounds — including brominated and fluorinated plastics typically used in fire-retardant applications — making Urban Rig uniquely suited to process waste streams that no other technology can safely handle.

Source: Urban Rig Recycling Solutions ENG, p.2 & p.14; Asada Shokai test report, 2022.

Four Marketable Oil Fractions

The 7-zone continuous process separates recovered oil into four distinct fractions by temperature-controlled condensation:

URC-2000 Zone	Temperature	Expected Yield	Oil Type	Flash Point	Density (SG)	Sulfur	Calorific Value	Possible Uses
Zone 1 (GET)	150°C	~ 10%	Heating Oil (Light naphtha), C5-C8	< 21°C	0.70-0.76	19-30 ppm	~44-46 MJ/kg	Petrochemical feed-stock (naphtha); Industrial solvent; Collected for resale
Zone 2 (GET)	300°C	~ 80%	Gas (Diesel) Oil (Kerosene + Diesel), C8-C20	40-55°C	0.80-0.86	50-150 ppm	43-47 MJ/kg	Recycled diesel fuel (NKKK certified); SAF co-processing feed-stock (Tier 1-3); JIS Class A/B heavy fuel oil equivalent; Industrial boiler combustion
Zone 3 (GET)	450°C	~ 10%	Heavy Oil, C20+	> 70°C	> 0.90	100-200 ppm	40-45 MJ/kg	Industrial heavy fuel oil; High-capacity boiler fuel; Bitumen/asphalt feed-stock

URC-2000 Zone	Temperature	Expected Oil Type	Flash Point	Density (SG)	Sulfur	Calorific Value	Possible Uses
Zone 4 (GET)	600°C	Traces	Asphaltenes/a (semi-solid residue)	> 1.0	Variable	~32 MJ/kg	Solid fuel (carbide); Soil amendment (biochar)

Note: Only the non-condensable gas fraction produced during pyrolysis is recycled to fuel the machine itself (after initial startup). All condensed oil fractions (Zones 1-4) are collected for commercial use or resale.

Sources by zone:

- **Zone 1 (~10%):** URC-2000 brochure; Asada Shokai report p.28 — lightest fraction of Class 1 (boiling point < 150°C); SG and sulfur from Class 1 PE distillation (0.756, 19 ppm).
- **Zone 2 (~80%):** URC-2000 brochure; Asada Shokai p.28 — Class 2+ (52-70%) combined with heavier Class 1 compounds (boiling point 150-300°C); Ohsumi Tab.3-8: flash < 40°C, 43-47 MJ/kg; NKKK certified test: flash ≤ 40°C, S 0.014%, 43.27 MJ/kg.
- **Zone 3 (~10%):** URC-2000 brochure; Asada Shokai — heavy tail of Class 2+ (boiling point > 300°C); Ohsumi Tab.3-9: comparable to JIS heavy fuel oil C.
- **Zone 4 (traces):** URC-2000 brochure; Ismitec lab analysis: urethane carbide 32.09 MJ/kg.

Data interpretation: The 10/80/10 distribution is a manufacturer specification for the continuous URC-2000 machine. It is supported by independent laboratory distillation tests conducted by Asada Shokai (Chiba Prefecture, 2022, report p.28), which separated crude batch oil into Japanese petroleum Class 1 (flash point < 21°C: 30-48% of crude) and Class 2+ (flash point > 21°C: 52-70% of crude). The apparent discrepancy between 30-48% Class 1 and the 10% light fraction is explained by the difference between flash point classification and condensation temperature separation: compounds classified as “Class 1” by flash point can have boiling points of 100-150°C+, meaning they would condense in Zone 2 (300°C) of the URC-2000, not Zone 1 (150°C). Only the lightest naphtha (boiling point < 150°C) is captured in Zone 1.

Underlying distillation data (Asada Shokai, 200 ml samples):

Plastic	Class 1 Vol.	Flash	SG	% Class 1	Class 2+ Vol.	Flash	SG	% Class 2+	Recovery
PE	61 ml	18°C	0.756	30.5%	139 ml	55°C	0.860	69.5%	100%
PP	56 ml	18°C	0.760	33%	114 ml	45°C	0.820	67%	85%
PS	90 ml	16°C	0.870	45%	110 ml	53°C	0.900	55%	100%
3P (mix)	96 ml	16°C	0.788	48%	104 ml	52.5°C	0.834	52%	100%

Source: Asada Shokai Industrial Waste Test Research Report (アサダショカイ工業株式会社), Section 7.2, p.28 (January 2023); URC-2000 product brochure; Ohsumi Verification Report, Table 3-8; NKKK certified test report.

At 90% resin content, total liquefied oil recovery reaches 76.5% of input (61.2% gas/diesel oil, 7.65% heating oil, 7.65% heavy oil). Oil quality has been certified by NKKK (Japan Marine

Survey Association): flashpoint 46°C, sulfur <0.2%, chlorine <100 ppm, with calorific value of 43.27 MJ/kg — comparable to fossil diesel at 42.6 MJ/kg.

Source: Urban Rig Recycling Solutions ENG, p.16-17; NKKK certified test report; UR2000-JP, p.4.

After initial startup, the non-condensable gas produced during pyrolysis is recycled to fuel the Urban Rig system itself, while all condensed oil fractions are collected for commercial use. Depending on quality grade, these oils can be resold on the market at a significantly more attractive price than the input waste, creating a strong revenue pathway. The only ongoing operational costs are labour and electricity.

Patented Thermal Transfer System

Urban Rig's patented thermal system delivers superior conductivity, condensation, and heat transfer performance in a highly controllable manner. The ultra-fine particle technology achieves combined spread, convective, and condensation heat transfer that exceeds conventional pyrolysis approaches. This results in finer temperature control and measurably better yields across all feedstock types.

Source: OneWorld Corporation patented technology.

Precision Automation — Minimal Staffing

The entire 7-process cycle — drying, heating, carbonization, oil extraction, and three-stage cooling — is automatically operated by microcomputer control from the loading slot through to the unloading slot. Three buckets move as one batch through seven zones every 90 minutes, enabling 24-hour continuous throughput of 48 batches (17 units of 3 buckets). This level of automation means that a 200 tonne/day plant can operate with minimal personnel, dramatically reducing operational costs compared to conventional waste processing facilities.

Source: UR2000-JP, p.4 & p.7; Urban Rig ENG, p.4.

Environmental Performance

Zero Dioxin and CO₂ Emissions

Because the reactor operates in a completely anoxic (oxygen-free) state, dioxins and carbon dioxide are simply not generated — unlike conventional incineration where oxygen reacts with carbon and chlorine compounds. This is a fundamental process advantage, not a filtration afterthought.

Source: Urban Rig Recycling Solutions ENG, p.14; UR2000-JP, p.4.

Emissions Far Below Regulatory Limits

Independent exhaust analysis by KANSO Technos confirms Urban Rig emissions are 5 to 50 times lower than French incineration standards:

Pollutant	Urban Rig Measured	French Incin. Limit	Factor Below Limit
HCl	<1 ppm	10 mg/m ³	>10x
SOx	<1 ppm	50 mg/m ³	>50x
NOx	15 ppm	200 mg/m ³	>13x
Particulates	<0.002 g/m ³	10 mg/m ³	>5x

Source: KANSO Technos independent exhaust gas analysis report.

Optional PRIMO Exhaust Upgrade

For deployments where even stricter air quality performance is desired, an optional PRIMO (PUREM 30.8) exhaust gas purification system is available. This module performs DeNO_x, DeSO_x, dust removal, and odour removal — using the same technology standard at thermal power plants worldwide — enabling Urban Rig to achieve near-zero atmospheric impact in sensitive urban or airport environments.

Source: Urban Rig Recycling Solutions ENG, p.12.

Integrated Water Purification (SEA CORAL)

The URC-2000 includes an integrated SEA CORAL ceramic purification system. This technology uses special ceramic as an absorbent with vaporisation heat to purify process water, extracting pure water free of minerals (including arsenic). It enables closed-loop water management within the facility, eliminating wastewater discharge.

Source: Urban Rig Recycling Solutions ENG, p.13; UR2000-JP technical documentation.

Low Noise and Odour

Thermal decomposition occurs entirely inside enclosed equipment, producing virtually no odour. Operating noise is comparable to that of a cooling tower, ensuring compatibility with nearby residential or commercial zones.

Source: UR2000-JP, p.3; Asada Shokai test report operational observations.

Safety

No Explosion Risk

Vaporised oil components are never oxidised because no oxygen is present in the reactor. Consequently, there is no risk of explosion or ignition at any point in the process. This represents a fundamental safety advantage over conventional thermal treatment methods.

Source: UR2000-JP, p.3.

Corrosion Protection for All Halogenated Compounds

The catalytic layer adsorbs chlorine from PVC processing, while steam neutralises hydrochloric acid that would otherwise corrode the reactor. The non-oxidising atmosphere further protects all equipment surfaces. Recent testing confirms this protection extends to all halogenated compounds (chlorinated, brominated, and fluorinated plastics — including those commonly used in fire-retardant applications), ensuring long equipment life even with challenging feedstock.

Source: Urban Rig Recycling Solutions ENG, p.14; UR2000-JP, p.3.

All Outputs Are Marketable

Every output from the Urban Rig process is a saleable resource: fuel oil (four fractions), industrial-grade charcoal (carbide), and recovered metals and glass. The carbide has a calorific value of approximately 25 MJ/kg — comparable to coal (23–30 MJ/kg) — and can be used as solid fuel, including to power the Urban Rig system itself. Metals are recovered in non-oxidised condition, yielding high-purity copper and other materials for direct recycling.

This contrasts sharply with conventional incineration, which produces toxic bottom ash (machefer - boiler slag) and fly ash requiring expensive hazardous landfilling. Urban Rig generates no such toxic residue — the entire output stream has positive market value.

Source: UR2000-JP, p.3; Annex A — Current Status Analysis; Urban Rig Recycling Solutions ENG, p.18.

Massive Volume Reduction

Urban Rig achieves carbonization reduction to approximately 25% of input volume (from roughly 12 tonnes input to 3 tonnes carbide). Critically, unlike incineration residue, the vast majority of this reduced volume is reusable: carbide serves as high-calorie solid fuel, metals and glass are directly recyclable. Even compared to incineration — which also reduces volume but produces toxic, non-reusable machefer (boiler slag) — Urban Rig delivers a far higher proportion of marketable outputs from the same reduction ratio.

Source: UR2000-JP, p.7; Urban Rig Recycling Solutions ENG, p.16.

Versatility — Processing Hard-to-Recycle Waste

Urban Rig's anoxic superheated steam process is uniquely capable of safely and profitably treating waste streams that are considered difficult or impossible to recycle through conventional mechanical or chemical means. This versatility opens entire market segments that existing waste technologies cannot serve.

Municipal & Household Waste

- General household waste (all components, unsorted)
- Food waste and organic residues (high moisture content)
- Mixed packaging and composite materials

Industrial & Manufacturing Waste

- Industrial plastic scrap — mixed polymer streams, production off-cuts, contaminated rejects that mechanical recyclers refuse
- Paint and coating waste — solvent-based paints, lacquers, varnishes, and dried paint residues containing heavy metals and volatile compounds
- Automotive shredder residue (ASR) — the mixed plastic, rubber, foam, and textile fraction remaining after metal extraction from end-of-life vehicles, typically landfilled due to its complexity
- Fibre-reinforced plastics (FRP/GRP) — glass-fibre and carbon-fibre composites from aerospace, marine, and construction sectors, which cannot be mechanically recycled
- Multi-layer packaging — laminated films combining aluminium, PE, and paper (e.g. Tetra Pak, coffee pouches) that defeat conventional recycling

Tyres & Rubber Waste

End-of-life tyres represent one of the world's most persistent waste challenges: vulcanised rubber cannot be melted or mechanically recycled, and their complex crosslinked structure combined with steel, textile, and chemical additives makes them extremely difficult to process. An estimated 1 billion waste tyres are generated globally each year. Urban Rig's anoxic thermal decomposition breaks down tyre waste into recoverable oil, carbon black (carbide), and steel — turning an environmental liability into three saleable resources.

Medical & Sanitary Waste

Healthcare facilities generate vast quantities of contaminated plastics — less than 5% of which are currently recycled. Urban Rig's high-temperature anoxic process provides complete thermal sanitisation, safely destroying pathogens, pharmaceutical residues, and biological contaminants while recovering oil and charcoal. Applicable waste streams include:

- Single-use medical devices, syringes, IV bags, and tubing
- Surgical gowns, drapes, and disposable PPE
- Pharmaceutical blister packs and contaminated packaging
- Laboratory waste and biological cultures (after autoclaving)

Sanitation-Regulated Waste (ICAO Category)

International regulations (notably ICAO and EU Regulation 1069/2009) require that aircraft cabin waste from international flights be sanitised or incinerated to prevent the spread of animal and plant diseases. This "catering waste" from aircraft — along with similar quarantine waste from ports and international logistics — currently goes straight to high-cost incineration with zero resource recovery. Urban Rig's high-temperature anoxic process meets sanitation requirements while simultaneously recovering valuable oil and charcoal from this waste stream.

Agricultural & Organic Waste

- Livestock waste — yielding animal-derived drainage and oil
- Agricultural film and silage wrap (contaminated LDPE that recyclers reject)
- Flower and crop waste
- Pesticide-contaminated containers and packaging

Other Hard-to-Recycle Streams

- Wire cord waste — with non-oxidised copper recovery (pure copper)
- Deposition film waste (aluminium/PE composites) — yielding mineral oil
- Textile waste, synthetic leather, and carpet (mixed-fibre materials that cannot be mechanically separated)
- Electronic waste plastics — brominated flame-retardant casings from WEEE (Waste Electrical and Electronic Equipment)
- Mattresses and furniture foam — polyurethane foam with fire retardants, a major landfill contributor
- Nappies and absorbent hygiene products — mixed plastic, cellulose, and superabsorbent polymers, representing up to 4% of household waste
- Marine and fishing waste — nets, ropes, and mixed ocean plastics contaminated with salt, biological matter, and UV degradation

Source: Urban Rig Recycling Solutions ENG, p.18; Urban Rig ENG, p.5; UR2000-JP; Waste Dive healthcare plastics report; EU Horizon Magazine on tyre pyrolysis.

FORCE 9 Companion Technology

For specialised applications, the FORCE 9 far-infrared catalytic reduction facility can detoxify incinerated ash and carbide containing heavy metals and harmful substances at approximately 400°C. Treated waste becomes usable as fertilisers, dehydrating agents, soil conditioners, and construction materials. This technology can even treat radioactively contaminated soil.

Source: Urban Rig Recycling Solutions ENG, p.11.

Scalability and Deployment

Urban Rig offers a complete product line from 1 tonne/day batch units to 200 tonne/day continuous plants:

Model	Capacity	Type	Footprint / Notes
URB-10	1 T/day	Batch (stationary)	Fits 40-foot marine container
URT-10	1 T/day	Batch (mobile)	Mounted on 10-tonne truck
URB-50	5 T/day	Batch	Small industrial site
URC-400	40 T/day	Continuous (7-stage)	2,700 m ² building
URC-1000	100 T/day	Continuous (7-stage)	Mid-scale municipal
URC-2000	200 T/day	Continuous (7-stage)	4,650 m ² building

Each processing unit (~12 tonnes) is factory mass-produced, enabling significant cost reduction. Units are split into 7 sections, transported by truck, and assembled on-site. Manufacturing lead time is approximately 10 months for continuous type and 6 months for batch type. Multiple URC-2000 plants can be combined — for example, 5x URC-2000 yields 1,000 tonnes/day capacity on a 28,900 m² site — enabling deployment at enterprise, industrial zone, or municipal scale.

Source: Urban Rig ENG, p.3 & p.19; UR2000-JP, p.7-10.

The Market

Addressable Feedstock Segments

Urban Rig’s anoxic steam cracking process is feedstock-agnostic — it profitably processes waste streams that conventional recycling and incineration handle poorly or not at all. The table below compares the most promising feedstock segments in France by volume, expected oil yield, current gate fee (what waste producers pay for disposal), and supply concentration — the ease with which feedstock can be reliably aggregated at a single plant location.

Each segment represents a distinct revenue opportunity. In practice, a plant at a site like CDG would operate on a blended feedstock portfolio, combining a baseload of locally concentrated waste (MSW, airport waste) with higher-value batches of industrial plastics or tyres when available.

Feedstock Segment	France Volume/yr	Oil Yield	Gate Fee	Supply Concentration
Industrial Pure Plastics	~2.2 MT	70-82%	€60-120/T	Excellent — industrial parks; consistent; contractable
MSW (residual)	~21.6 MT	20-66%	€80-180/T	Excellent — daily municipal collection; enormous volume
End-of-Life Tyres	~530,000 T	40-50%	€0-50/T	Moderate — ÎdF ~95,000 T in ~5 weeks; batch not baseload
Aircraft & Quarantine	~80-100,000 T (CDG)	25-40%	€120-200/T	Excellent at CDG — single site; captive volume
Medical / DASRI	~700,000 T total	30-50%	€800-1,500/T	Good — hospitals in urban areas; ÎdF highest density
Automotive Shredder Residue	~200-350,000 T	35-50%	€100-200/T	Good — ~40 shredder sites; large volumes per site
Composite & Multi-Layer	~500,000 T+	40-60%	€80-150/T	Moderate — via sorting centres; EPR partnerships

Source: ADEME 2022; Aliapur annual report; ETRMA; Eurostat industrial waste statistics; Annex A & B.

Industrial pure plastics offer the highest oil yield and the most attractive economics per tonne, but volumes are finite and already partially contested by mechanical recyclers. MSW represents the largest addressable volume by far and provides a reliable, locally abundant baseload with strong gate fees. End-of-life tyres yield good oil quality with higher calorific value, but their national collection infrastructure means that a single plant’s regional catchment would exhaust its tyre supply rapidly — making tyres an excellent supplementary batch feedstock rather than a primary baseload. Medical waste commands the highest gate fees (up to €1,500/T for DASRI) but volumes are smaller. The optimal strategy is a blended portfolio: MSW or airport waste as baseload, supplemented by high-margin batches of industrial plastics, tyres, and medical waste.

Total Addressable Market

Global MSW generation stands at 2.01 billion tonnes/year (2016), projected to reach 3.4 billion by 2050. In Europe alone, 225 million tonnes are generated annually. The EU Circular Economy Package mandates 65% recycling by 2035 and maximum 10% landfilling — creating urgent demand for technologies that can process mixed residual waste into valuable resources.

Source: World Bank “What a Waste 2.0” (2018); Eurostat (2023).

France — Initial Target Market

France generates 38.5 million tonnes of MSW annually. Of this, 38.1% is incinerated and 18.1% landfilled — representing approximately 21.6 million tonnes per year of waste that could be redirected to resource recovery via Urban Rig technology.

French tipping fees provide strong economic incentive: landfilling costs €140–180/tonne, incineration €80–120/tonne. Urban Rig's ability to generate revenue from waste outputs (oil, charcoal, metals) while also collecting tipping fees creates a dual revenue model that fundamentally changes waste economics.

Source: ADEME 2022; Annex B — France Market Analysis.

Île-de-France and CDG Airport

The Île-de-France region (12.3 million population) generates 6.2 million tonnes/year. Paris-Charles de Gaulle Airport alone produces an estimated 80,000–100,000 tonnes/year of mixed waste from aircraft cabins (40%), terminals (30%), catering (15%), and cargo logistics (10%).

Airport waste is particularly suitable for Urban Rig processing: it contains a high proportion of plastics and packaging, is generated at a single concentrated location, and airports face increasing pressure to demonstrate sustainability credentials. A single URC-2000 unit (200 T/day nominal, ~140 T/day actual weight) can process the airport's entire waste stream with capacity to serve surrounding communities.

Source: Annex B — CDG Airport waste estimates based on 76.2M passengers (2019).

The Business Model

Revenue Streams

Urban Rig generates four distinct revenue streams from a single waste input. The following estimates are based on the URC-2000 reference deployment (200 T/day nominal, 140 T/day actual weight, 46,200 T/year at 330 operating days) and are configurable to any site and feedstock strategy:

Revenue Stream	Illustrative Annual Estimate	% of Total
Gate fees (waste intake)	€5.3M	33%
Fuel oil sales (three fractions)	€6.3M	39%
Char/biochar sales	€1.4M	9%
Recovered metals & recyclables	€0.6M	4%
Carbon credits (CO ₂ avoided)	€2.5M	15%
TOTAL	€16.1M	100%

Source: UR-Financial-Model-Restructured-V3.xlsx (Year 3, full capacity). Throughput: 200 T/day volumetric × 0.7 bulk density = 140 T/day actual × 330 days = 46,200 T/year.

Oil Pricing

Oil pricing has been modelled at three scenarios based on current market benchmarks (ICIS plastic pyrolysis oil index, NWE naphtha, fossil gasoil):

Scenario	Weighted Price (€/T)	Price per Litre	Annual Oil Revenue (Year 3)
Conservative (Low)	€480/T	€0.41/L	~€4.7M
Base Case (Recommended)	€641/T	€0.55/L	~€6.3M
Optimistic (ISCC premium)	€895/T	€0.76/L	~€8.8M

Source: Annex G — Oil Fraction Pricing Model; ICIS (2024–2025).

Importantly, Urban Rig’s output oil is derived from waste-to-resource conversion rather than fossil extraction, qualifying it under EU sustainability frameworks. Because the feedstock is pre-sorted waste that would otherwise be incinerated or landfilled, the recovered oil carries a favourable sustainability classification. Under schemes such as ISCC (International Sustainability and Carbon Certification), this waste-derived oil commands a significant premium over conventional fossil equivalents — reflected in the optimistic scenario above. As circular economy regulations tighten across Europe, this premium is expected to widen further, making Urban Rig oil increasingly valuable relative to fossil benchmarks.

Source: ISCC certification framework; ICIS Plastic Pyrolysis Oil index; EU Renewable Energy Directive (RED II/III).

Illustrative Financial Summary (France, URC-2000)

Metric	Value (Year 3, full capacity)
CAPEX	€63.8M
Annual throughput	46,200 T/year (140 T/day × 330 days)
Annual Revenue	€16.1M
Annual OPEX	€7.3M
EBITDA	€8.9M (55% margin)
Simple Payback	~7 years (CAPEX ÷ EBITDA)
Equity Payback	~12 years
Debt-service cash flow breakeven	Year 6

Source: UR-Financial-Model-Restructured-V3.xlsx. OPEX includes purchased heating oil, pre/post-sorting operations, and ash disposal. Financing: 70% debt at 12% over 10 years, 30% equity.

Energy Supply Model

The V2 model uses **purchased heavy fuel oil for heating**, ensuring reliable and consistent operations independent of feedstock composition.

Heating supply: - Heating demand: 2,714 MJ per tonne processed - Annual cost: ~€1.0M at full capacity (€0.40/L, HFO Rotterdam benchmark) - Ensures steady-state operations and predictable costs

By-product value: - **Char/biochar** (~41.4 T/day at full capacity, 13,673 T/year): All char sold commercially at EUR 100/T (fuel briquettes) or EUR 350/T (certified agricultural biochar) - Annual revenue: €1.4M (briquette scenario, used in base case) to €4.8M (agricultural biochar scenario) - **Off-gas:** Monitored for composition and energy content (provides valuable data

on process efficiency) - Not used for internal heating (off-gas recovery too variable) - Energy content helps understand commercial product value

Rationale: This approach maximizes product revenue while ensuring reliable, predictable operations. The off-gas and char energy data are valuable for understanding process efficiency and product quality, but are not required for heating supply.

Source: UR-Financial-Model-Restructured-V3.xlsx; Annex A §A.8; Annex F (char product analysis).

Reference Deployment Model: CDG Airport

To illustrate how a deployment is structured, we present Paris-Charles de Gaulle Airport as a reference model. CDG was selected because it exemplifies the characteristics of a strong candidate site: concentrated waste flows from multiple sources, high-value feedstock mix (plastics-rich cabin and packaging waste), premium gate fees (ICAO-regulated quarantine waste), proximity to industrial offtake markets, and a site operator under sustainability pressure. The following analysis demonstrates the methodology an investor would apply to evaluate any candidate location. A single URC-2000 unit (200 tonnes/day continuous) would process mixed airport waste and surrounding regional MSW.

Feedstock Analysis

French MSW composition processed by Urban Rig, after full decomposition of all 13 MODECOM categories into constituent materials:

Component	% of Processable Feedstock	Primary Output
Food & Organic	38.4%	Charcoal (carbide)
Paper, Card, Wood & Cotton	37.9%	Charcoal + off-gas
Plastics & Synthetic Polymers	23.7%	Fuel oil (4 fractions)

Source: Annex F — MSW Feedstock Yield Model (ADEME MODECOM 2017 full 13-category decomposition). Processable feedstock = 859.1 kg per tonne after removing 109.5 kg of glass, metals, incombustibles, and fragments, and excluding 31.4 kg mineral inert.

Yield Scenarios

Scenario	Description	Oil Yield
A: Full OMR (with decomposition)	Mixed MSW as received	23%
B: Plastic-like fraction	Pre-sorted plastic-rich stream	65%
C: PE+PP polyolefins only	High-value polyolefin stream	78%

Source: Annex F; Asada Shokai test data (9 resin types).

Waste Composition — CDG Airport

Charles de Gaulle Airport generates an estimated 80,000–100,000 tonnes/year of mixed waste:

- Aircraft cabin waste: 40% (32,000–40,000 T) — subject to ICAO sanitation regulations
- Terminal waste: 30% (24,000–30,000 T)
- Catering waste: 15% (12,000–15,000 T) — quarantine waste requiring thermal treatment
- Cargo and logistics: 10% (8,000–10,000 T)
- Other: 5%

Source: Annex B estimates based on 76.2M passengers (2019), 1.2–1.5 kg waste/passenger.

Deployment Economics & Partnership Model

The following illustrates the capital requirements and return profile for a URC-2000 European deployment (200 T/day nominal, 46,200 T/year actual throughput), using the CDG reference model. These figures are configurable: site selection, feedstock strategy, and scale are determined jointly with the investment partner. The economics below represent one scenario output of the financial model.

Illustrative Capital Requirements (URC-2000)

Item	Estimated Cost
URC-2000 equipment (200 T/day nominal)	€50.0M
Building and site preparation	€5.0M
Installation, freight, and training	€0.9M
Accessories, spares, and monitoring	€2.0M
Pre-sorting equipment	€1.1M
Post-sorting equipment	€0.4M
Compliance and engineering (ATEX, ICPE, detailed engineering, project management)	€1.4M
Contingency (5%)	€3.0M
TOTAL	€63.8M

Financing structure: 70% bank debt (€44.6M at 12% over 10 years) / 30% equity (€19.1M).

Illustrative Return Profile

- EBITDA: €8.9M/year (55% margin) at full capacity — heating oil cost built into OPEX; char sold commercially
- Simple payback: ~7 years (CAPEX ÷ steady-state EBITDA)
- Equity payback: ~12 years (cumulative free cash flow net of equity investment)
- Debt-service cash flow breakeven: Year 6 (cumulative cash flow after debt service turns positive)
- Diversified revenue: gate fees (€5.3M) + oil (€6.3M) + char, metals, carbon credits (€4.5M)
- Annualised ROI on equity: ~17.6%
- Scalable: each additional URC-2000 replicates the model

Source: UR-Financial-Model-Restructured-V3.xlsx (Year 3 = first full-capacity year; 15-year projection with 2% annual inflation).

Proven Technology

Urban Rig technology is validated by:

- Asada Shokai industrial test programme (Noda factory, Chiba Prefecture, 2022) with independent measurement by Land Brain Co., Ltd.
- KANSO Technos independent exhaust gas analysis: emissions 5–50x below French incineration standards
- NKKK (Japan Marine Survey Association) certified oil quality testing
- Field-proven operations: processing plant in Shandong Qufu (China) using recovered diesel oil as operational fuel
- Sea Waste Zero Award — Innovation category, Japan Foundation / Ministry of the Environment (June 2019)
- “Plastics Smart” designation — Ministry of Economy, Trade and Industry, Japan

Source: Asada Shokai report; KANSO Technos report; NKKK certification; OneWorld Corporation records.